



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	BIOPHOTONICS				
Type of Course	Major Elective (SPECIALIZATION II: PHOTONICS)				
Semester	VI				
Academic Level	300 – 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Fundamental knowledge in optics, photonics and biology				
Course Summary	Biophotonics is an interdisciplinary field that combines principles of physics, biology, and optics to study biological systems using light-based techniques. This course covers topics such as optical properties of biological tissues, imaging and biosensing techniques, instrumentation, and emerging trends. Students gain both theoretical knowledge and practical skills through lectures, and projects/lab, preparing them for careers in research, healthcare, and technology development.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding photobiology contributes to advancements in medical treatments, such as photodynamic therapy using exogenous photosensitizers.	U	C	Written exams, quizzes

CO2	Imaging in the field of biophotonics provides a comprehensive understanding of visualization techniques at various scales within the biological system.	An	p	Quizzes, presentations
CO3	Studying the principles of optical biosensing, equips individuals with the knowledge to design, develop, and apply advanced sensing technologies.	U&Ap	C	Written exams, experiments
CO4	Understanding the techniques of a flow cytometer, tweezers, optical responses, and the principles of photodynamic therapy fosters the development of advanced diagnostic and therapeutic techniques. Additionally, exploring This knowledge contributes to advancements in both clinical diagnostics and biological research..	U & Ap	C	Written exams, quizzes
CO5	Apply Photonics principles to real-world applications such as imaging and sensors	Ap	C	Mini Projects
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge (F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Photobiology		14	20
	1	Photobiology—At the Core of Biophotonics , Interaction of Light with Cells - Light Absorption in Cells, Light-Induced Cellular Processes,	3	
	2	Photochemistry Induced by Exogenous Photosensitizers ,Interaction of Light with Tissues - Photoprocesses in Biopolymers	3	
	3	Human Eye and Vision , Photosynthesis	2	

	4	In Vivo Photoexcitation - Free-Space Propagation	1	
	5	Optical Fiber Delivery System , Articulated Arm Delivery, Hollow Tube Waveguides,,	2	
	6	In Vivo Spectroscopy	1	
	7	Optical Biopsy ,Single-Molecule Detection,	2	
		Chapter 6 of Book 1		
II	Bioimaging: Principles and Techniques		11	15
	8	An Overview of Optical Imaging	1	
	9	Transmission Microscopy - Simple Microscope, Compound Microscope, Kohler Illumination	3	
	10	Fluorescence Microscopy, Confocal microscopoy,	2	
	11	Fluorescence Resonance Energy Transfer (FRET)Imaging ,Fluorophores as Bioimaging Probes	2	
	12	Imaging of Organelles, Imaging of Microbes, Cellular Imaging, Tissue, <i>In Vivo</i> Imaging	3	
		Chapter 7 of Book 1		
III	Optical Biosensors		9	10
	13	Principles of Optical Biosensing	2	
	14	Fiber Optic Biosensors, Evanescent Wave Biosensors,	2	
	15	Surface Plasmon Resonance Biosensors	2	
	16	Some Recent Novel Sensing Methods, Commercially Available Biosensors.	3	
		Chapter 9 of Book 1		
IV	Photonic tools for medical applications		14	25
	17	Flow Cytometry A Clinical, Biodetection, and Research Tool,	2	
	18	Basics of Flow Cytometry- Basic Steps, The Components of a Flow Cytometer, Optical Response	3	
	19	Photodynamic Therapy: Basic Principles,	1	

	20	Laser Tweezers and Laser Scissors- New Biological Tools for Micromanipulation by Light Principle of Laser Tweezer Action	3	
	21	Design of a Laser Tweezer 490 Optical Trapping Using Non-Gaussian Beams	3	
	22	Laser Scissors -Laser Pressure Catapulting (LPC), Laser Capture Microdissection (LCM),	2	
		Sections 11.1,11.2, 12.1,14.1, 14.2, 14.3, 14.4, 14.6 of Book 1		
V	Open Ended Module: Mini project		12	
		1. Synthesis of Semiconductor Quantum Dots for Bioimaging		
		2. Fabrication of biosensors using optical fibers		
Text Book for study 1. Introduction to Biophotonics, Paras N Prasad, Wiley Interscience, A John Wiley & Sons, INC Publication (Book 1) References: 1. Biomedical Photonics –A handbook-T.Vo Dinh (CRC Press 2002) 2. Nanophotonics, Paras N Prasad (Wiley Interscience, 2003) 3. Optic Fiber Communications, Gerd Keiser (McGraw –Hill International Editions)				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PS O4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	0	0	0	0	0	3	2	0	0	0	0	3
CO 2	3	2	0	0	0	0	3	2	0	0	0	0	3
CO 3	3	0	0	0	0	0	3	2	0	0	0	0	3
CO 4	3	0	0	3	0	0	3	2	0	2	0	0	3
CO 5	3	0	0	0	0	0	3	2	0	0	0	0	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓